

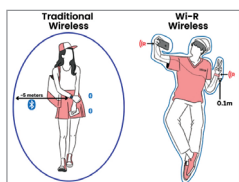
TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Non-radiative wireless solution

Ixana's Wi-R solution utilise non-radiative wireless signal to transfer data. It consumes up to 100 times lower energy compared to the existing Bluetooth



and Wi-Fi protocols. Ixana offers Wi-R solution through YR11, YR22, and YR33 chips

that offer a speed of up to 1Mbit/s, 4Mbit/s, and 20Mbit/s, respectively. The high-speed, low-powered solution enables distributed AI with a wireless hub. This technology utilises a human body as a conductor and confines the signal around the user's body. It can allow engineers to design devices that provide robustness, efficiency, and touch selectivity of the signal. It also enhances safety of the system by having a short transmission range from the body and featuring mathematical security that comes from encryption. The Wi-R solution is suitable for use in wearables and audio devices like wireless headphones and smart watches.

Ixana.ai

<https://ixana.ai>

AC direct DC power IC

Amber Semiconductor's AC direct DC power delivery IC is capable of extracting DC directly from AC mains without the use of transformers, rectifier bridges, or filtering. The AC direct DC enabler silicon chip enables dynamic delivery of DC power on demand. It requires lesser components for achieving power conversion, thus



reducing the size, cost, and complexity for designing a power converter. It can take a wide range of input from 25 to 277V AC to provide a DC output of 1.8V to 24V. AC direct DC power delivery will enable engineers to develop low-cost, compact power supplies or products running on low-voltage DC. It is suited for a broad range of electrical products, such as smoke detectors, doorbell cameras, thermostats, smart products, appliances, and intelligent HVACR.

Amber Semi

<https://ambersi.com>

High-efficiency application processor

R2 FastLane Core Power is an application processor by R2 Semiconductors that can improve overall system performance. The application processor offers higher efficiency by reducing heat generation and supports both fast changing load voltages and currents. It is based on a high switching frequency, high bandwidth technology, which enables dynamic voltage and frequency scaling to adjust power supply voltage, match load conditions, reduce wasted power consumption and heat. R2 will enable system on chip developers to integrate efficient power management, thereby reducing form factor and noise interference. It will help design devices with improved system performance, smaller footprint, better thermal management, and reduced power consumption, thus, reducing the size and cost of the product.

R2 Semiconductors

<https://r2semi.com>

Moisture and temperature sensor

The AZN3100 from Axzon is a flexible, wireless, battery-free moisture

and temperature sensor. It is designed to monitor environmental and material moisture and temperature content over a wide range of settings. The sensor has a read range of up to 8 metres and is



optimised for direct contact with moisture or water droplets. The

AZN3100 incorporates an integrated RF antenna to harvest energy for the moisture sensing function, as well as to communicate with a RAIN/UHF compliant reader. The integrated antenna, configured to be sensitive to moisture and rain, is built using flexible materials with a built-in adhesive strip. The sensor can be used to detect moisture and temperature in applications that include bulk material like grain, food products, as well as building and roofing materials.

Axzon

<https://www.axzon.com>

Photoelectric sensor front-end

Triad Semiconductor's TS8003 is a highly integrated photoelectric sensor analogue front-end IC. It combines an optical source modulated trans-



mit driver, dual high-performance photodiode receive channels, high input voltage

DC-DC converter/linear regulators, and user configurable high-voltage IO-Link/binary output stages in a single chip. The TS8003 comes in a compact 30-bump wafer level chip scale package (WLCSP, 2.6mm × 3mm, 0.5mm pitch). It can operate over 9 to 36V nominal supply input and over -25 to +85°C operating temperature range. The TS8003 provides a complete mixed-signal interface solution for multiple